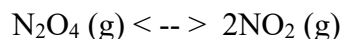


Chapter 16 (Equilibrium), Chapter 17 (Acids & Bases)

- 1) Understand how a catalyst affects the equilibrium constant for a reaction
- 2) Consider the following reaction:



The K_c for the reaction is 5×10^{-7} . If 30 moles of the N_2O_4 are placed in a 60 liter container, what are the final concentrations of the reactant and products? Hint: start by calculating the initial concentration of N_2O_4

- 3) What is meant by dynamic equilibrium?
- 4) Given the equilibrium constant for a reaction, calculate the equilibrium constant for the reverse reaction.
- 5) Understand LeChateliers affect. I.e. if a reaction is at equilibrium and a product or reactant is added or removed, how if the reaction affected. i.e. does it move to the left, right, or is it unchanged.
- 6) Explain with words, symbols or chemical reactions why an aqueous solution of CN^- can be used to dissolve gold. The K_{eq} for $\text{Au}^+ + 2\text{CN}^- < \text{--} > \text{Au}(\text{CN})_2^-$ is 2×10^{38} . The K for the reaction $\text{Au} < \text{--} > \text{Au}^+$ is 1×10^{-20}
- 7) Given the K_c or K_p for a reaction, calculate the other. i.e. similar to Qs 33 & 34.
- 8) Identify the conjugate acid or conjugate base formed from an acid base reaction
- 9) Know the definition of an Arrhenius acid/base, conjugate acid/base, and Lewis acid/base
- 10) Understand the relationship between pH and POH. Understand how to calculate the pH or pOH of a strong acid/base.
- 11) Given a weak acid or base, calculate the pH or pOH.
- 12) Understand the dampening effect of water as it relates to the strongest acid or base that can exist in water.
- 13) Using inductive or resonance arguments, predict the relative strength of two (or more acids).
- 14) Calculate the percent ionization of a solution of a weak acid or base.
- 15) Identify which cations will create an acidic solution.
- 16) Given the concentration of a weak acid/base and its pH, calculate the K_a/K_b
- 17) Given the K_a/K_b or an acid/base, determine the K_b/K_a of the conjugate base/acid

Spring 2025 (West Valley)

Chapter 18 (Solubility, K_{sp} , etc), Chapter 19 (Thermodynamics), Chapter 20 (Electrochemistry)

All questions worth 6 pts unless noted otherwise. Show all work for full credit.

Which of the following statements best describes the second law of thermodynamics?

- Energy is conserved.
- Entropy is conserved
- Entropy continually increases
- Entropy continually decreases
- Energy continually increases
- Energy continually decreases

Which of the following is expected to have the Lowest entropy?

- Solids
- Liquids
- Gases
- insufficient information

3) Magnesium hydroxide dissolves in water to the extent of 1.3 g per liter. What is the value of K_{sp} for $Mg(OH)_2$? (10 pts)

4) If you mix 48 mL of 0.0012 M $BaCl_2$ with 24 mL of 1×10^{-6} M Na_2SO_4 will a precipitate of $BaSO_4$ form? The K_{sp} for $BaSO_4$ is 1.1×10^{-10} . (12 pts)

5) Heating aluminum carbonate leads to its decomposition. (12 pts)



Calculate ΔH and ΔS for the reaction.

Calculate ΔG under standard conditions

Above what temperature does the reaction become spontaneous? i.e., at what temperature does aluminum carbonate decompose.

6) The conversion of diamond to graphite is energetically favored. i.e. the graphite is more stable. However, any diamond that existed 100,000 years ago remains effectively unchanged today. Which statement is the best description for this situation.

Diamond is an allotrope of carbon that does not decompose.

Kinetics tells us that the reaction will occur. However, thermodynamics tells us how fast the reaction will occur

If the temperature and pressure are increased, the diamond will convert to graphite quickly.

Thermodynamics tells us that the reaction will occur. However, kinetics tells us how fast the reaction will occur.

All of the above

None of the above

7)The K_{sp} for $AgClO_4$ is 3.2×10^{-5} . What is the solubility of $AgClO_4$ in water? What is the solubility of $AgClO_4$ in 1.0 M $NaClO_4$? (10pts)

8)All spontaneous reactions are endothermic. (Circle the correct answer)

True, False, insufficient information

9)Write the ion product expression for iron (III) sulfide.

10)How does the molar entropy of a substance change with increasing temperature?

It increases

It decreases

Insufficient information provided

None of the above

11)Circle whether the sign of ΔS_{sys} for each process is positive or negative. This is not a multiple choice question.

Water boiling + -

Water freezing + -

Sublimation of water + -

12) Consider the reaction:



$$K_p = 2.26 \times 10^4.$$

Calculate ΔG_{rxn} under standard conditions. (8pts)

13) Balance the following oxidation reduction reaction in acid conditions.



14) Consider the voltaic cell below (problem 47 chapter 20)

Label the anode and cathode (2pts)

Draw an arrow for the direction and location of electron flow. (2pts)

Write an equation for the overall reaction (2pts)

Calculate E°_{cell} . (4pts)

Indicate the direction & location of anion and cation flow. (2pts)

15) Which metal is the best oxidizing agent.

Na^+ , Al^{3+} , Ag^+ or Fe^{2+}

16) The E° for a voltaic cell with the following reaction is 1.56 V. What is the equilibrium constant? (8pts)



17) A voltaic cell consists of Zn/Zn^{2+} half-cell and Ni/Ni^{2+} half cell. The initial concentrations of Ni^{2+} and Zn^{2+} are 1.50M and 0.100 M respectively. What is the initial cell potential? (8pts)

18) Which of the following are methods of prevention corrosion?

The use of a sacrificial anode

Painting

Galvanic protection

All of the above

None of the above

19) An electric eel generates an electric shock utilizing cells that act as concentration cells. The potassium ion concentration on the inside of the cell is 0.10 M and the concentration outside of the cell is 0.01M. What is the voltage of a single cell?

20) An electric eel can generate a total voltage of up to 600 V. However, the voltage generated by a single cell is 0.20 V. How is it possible to generate 600 V?

The cells are connected in parallel

The eel has millions of cells

The cells are connected in series.

The cells constantly regenerate

All of the above

None of the above

21) An electrolytic cell operates at 20 amps for 10 minutes. How many grams of Au^{3+} can be electroplated by this cell?